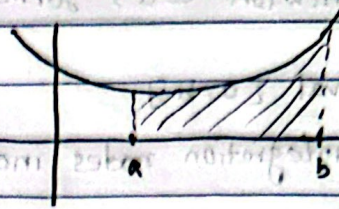


7.1

Definite Integral;

$$I(f) = \int_a^b f(x) dx$$

Means the exact area under the curve  $f(x)$  from  $a$  to  $b$ .



There is no exact formula for integral. So, we need an approximation.

$$I(f) = \int_a^b f(x) dx$$

↳ This is complex Realworld function

We approximate  $f(x)$  using a polynomial  $P_n(x)$

$$I(f) = \int_a^b f(x) dx \approx \int_a^b P_n(x) dx$$

We know,

$$\text{Lagrange Polynomial, } P_n(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + \dots + l_n(x)f(x_n)$$

$$= \sum_{k=0}^n l_k(x)f(x_k)$$

$$\text{So, } \int_a^b P_n(x) dx = \int_a^b \sum_{k=0}^n l_k(x)f(x_k) dx$$

$$\Rightarrow I_n = \sum_{k=0}^n f(x_k) \int_a^b l_k(x) dx$$

$$\therefore \sigma_k = \int_a^b l_k(x) dx$$

↳ weight function  
defined as  $\sigma_k$

$$\Rightarrow I_n = \sum_{k=0}^n f(x_k) \sigma_k$$

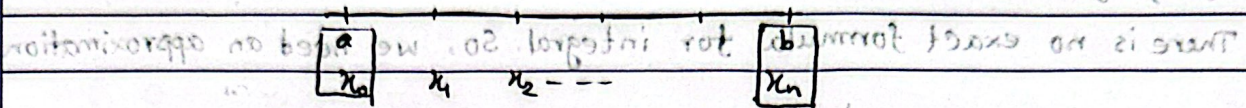
↳ General Quadrature Rule

If the nodes  $x_k$  are equally spaced in  $[a, b]$  then the formula becomes Newton's Cotes family (which includes trapezoidal Rule, Simpson Rule etc)



Closed Newton Cotes formulaInterval  $[a, b]$ 

The integration nodes includes the endpoints

 $\therefore$  nodes are,

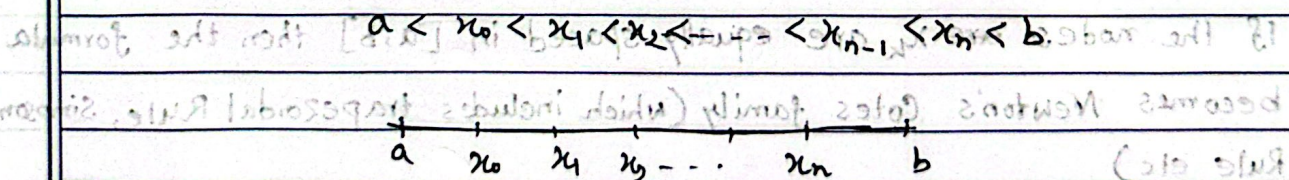
$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$$

Since nodes are equally spaced, the spacing

$$h = \frac{b-a}{n}$$

For  $n=1$ , we get Trapezium RuleFor  $n=2$ , we get Simpson RuleOpen Newton Cotes formulaInterval  $[a, b]$ 

The integration nodes does not include the endpoints.

 $\therefore$  Nodes are,

Since nodes are equally spaced, the spacing,

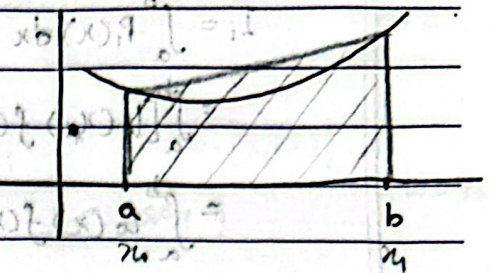
$$h = \frac{b-a}{n+2}$$



# Trapezium Rule: Closed Newton Cotes formula for degree $n=1$

Since,  $n=1$ , there will be 2 nodes.

As  $x \in [a, b]$   $h = \frac{b-a}{n} = \frac{b-a}{1}$   
 $\downarrow \quad \downarrow$   
 $x_0 \quad x_1$   $\therefore h = b-a$



$$P_1(x) = l_0(x)f(x_0) + l_1(x)f(x_1)$$

$$l_0(x) = \frac{x-x_1}{x_0-x_1} = \frac{x-b}{a-b}$$

$$l_1(x) = \frac{x-x_0}{x_1-x_0} = \frac{x-a}{b-a}$$

$$\omega_0 = \int_a^b f(x_0) l_0(x) dx$$

$$= \int_a^b \frac{x-b}{a-b} dx$$

$$= \frac{1}{a-b} \int_a^b (x-b) dx$$

$$= \frac{1}{a-b} \left[ \frac{x^2}{2} - bx \right]_a^b$$

$$= \frac{1}{a-b} \left[ \frac{b^2}{2} - b^2 - \left( \frac{a^2}{2} - ab \right) \right]$$

$$= \frac{1}{a-b} \left[ -\frac{b^2}{2} - \frac{a^2 - 2ab}{2} \right]$$

$$= \frac{1}{a-b} \left[ \frac{-(a^2 - 2ab + b^2)}{2} \right]$$

$$= \frac{1}{(a-b)} \frac{-(a-b)^2}{2}$$

$$= \frac{-(a-b)}{2} = -\frac{(b-a)}{2}$$

$$\therefore \omega_0 = \frac{b-a}{2}$$

$$\omega_1 = \int_a^b l_1(x) dx$$

$$= \int_a^b \frac{x-a}{b-a} dx$$

$$= \frac{1}{b-a} \int_a^b (x-a) dx$$

$$= \frac{1}{b-a} \left[ \frac{x^2}{2} - ax \right]_a^b$$

$$= \frac{1}{b-a} \left[ \frac{b^2}{2} - ab - \left( \frac{a^2}{2} - a^2 \right) \right]$$

$$= \frac{1}{b-a} \left[ \frac{b^2 - 2ab + a^2}{2} \right]$$

$$= \frac{1}{(b-a)} \frac{(b-a)^2}{2}$$

$$\therefore \omega_1 = \frac{b-a}{2}$$



We know,  $I_n = \int_a^b P_n(x) dx$

$$I_1 = \int_a^b P_1(x) dx$$

$$= \int_a^b [l_0(x)f(x_0) + l_1(x)f(x_1)] dx$$

$$= \int_a^b l_0(x)f(x_0) dx + \int_a^b l_1(x)f(x_1) dx$$

$$= f(x_0) \int_a^b l_0(x) dx + f(x_1) \int_a^b l_1(x) dx$$

$$= f(x_0) \omega_0 + f(x_1) \omega_1$$

$$= f(x_0) \frac{b-a}{2} + f(x_1) \frac{b-a}{2}$$

$$= \frac{b-a}{2} [f(x_0) + f(x_1)]$$

$$\boxed{I_1 = \frac{b-a}{2} [f(a) + f(b)]} \rightarrow \text{Trapezium Rule Formula}$$

Example: Given that,  $f(x) = e^x$  and interval  $[0, 2]$

(a) Find Numerical Integration using Trapezium Rule.

(b) Find actual Integration value.

(c) Find relative percentage error.

$\Rightarrow$  (a) We know,

$$I_1 = \frac{b-a}{2} [f(a) + f(b)]$$

$$= \frac{2-0}{2} [e^0 + e^2]$$

$$= 8.3891.$$



(b)  $\int_0^2 e^x dx$

$$= [e^x]_0^2$$

$$= e^2 - e^0$$

$$= 6.3891$$

(c) Relative Percentage Error

$$= \frac{| \text{Actual} - \text{Approximate} |}{| \text{Actual} |} \times 100\%$$

$$= \frac{| 6.3891 - 8.3891 |}{| 6.3891 |} \times 100\%$$

$$= \frac{2.0}{6.3891} \times 100\%$$

$$= 31.65\%$$

Upper Bound of Interpolation Error:

Cauchy's Error,

$$\text{Upper Bound Error} = \left| \frac{f^{(n+1)}(\xi)}{(n+1)!} \right| \left| \int_a^b (x-x_0)(x-x_1)\dots(x-x_n) dx \right|$$

From previous Example:

$n=1$ ,  $f(x) = e^x$  interval  $[0, 2]$ , find the max possible error for numerical interpolation.

$$\text{maximum error} = \left| \frac{f''(\xi)}{2!} \right| \left| \int_a^b (x-x_0)(x-x_1) dx \right|$$

$$\left| \frac{f''(\xi)}{2!} \right| = \left| \frac{e^x}{2!} \right|$$

$$\text{for max} = \left| \frac{e^2}{2!} \right| = \frac{e^2}{2}$$

$$= \left| \frac{8}{3} - 4 - 0 - 0 \right|$$

$$= \left| -\frac{4}{3} \right| = \frac{4}{3}$$

$$\therefore \text{Upper bound error} = \frac{e^2}{2} \times \frac{4}{3}$$

$$\int_a^b \left| \int_0^2 (x-0)(x-2) dx \right|$$

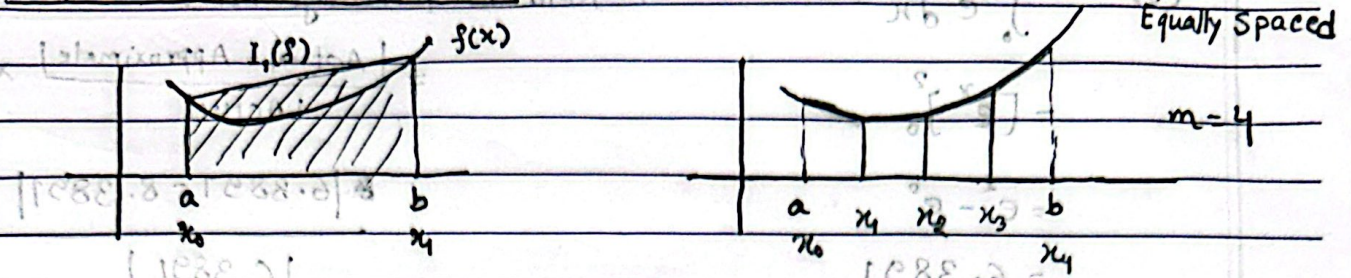
$$= \left| \int_0^2 (x^2 - 2x) dx \right|$$

$$= \left| \left[ \frac{x^3}{3} - \frac{2x^2}{2} \right]_0^2 \right|$$

$$= \left[ \frac{8}{3} - 4 \right] = -\frac{4}{3}$$

(Ans)



Composite Newton Cotes Formula:

Instead of doing this,

⊛ This method improves result without increasing the degree  $n$ .

Here we divide  $[a, b]$  into  $m$  subintervals with equal length.

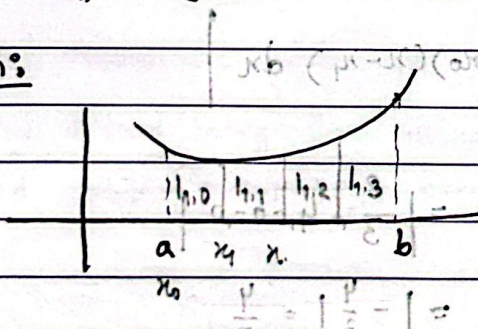
Space between Nodes,  $h = \frac{b-a}{m}$

We denote Composite Newton Cotes formula as  $C_{1,m}$

degree 1      no. of subinterval

$$C_{1,m} = \sum_{i=0}^{m-1} I_{1,i}$$

$$C_{1,m} = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{m-1}) + f(x_m)]$$

Derivation:

We know,

$$\text{Trapezium Rule} = \frac{b-a}{2} [f(a) + f(b)]$$

$$C_{1,4} = I_{1,0} + I_{1,1} + I_{1,2} + I_{1,3}$$

$$= \sum_{i=0}^3 I_{1,i}$$

$$h = \frac{b-a}{m}$$

$$= x_1 - x_0$$

$$= x_2 - x_1$$

$$= x_3 - x_2$$

$$= x_{i+1} - x_i$$



$$C_{1,m} = \sum_{i=0}^{m-1} I_{1,i}$$

Now,

$$I_{1,0} = \frac{x_1 - x_0}{2} (f(x_0) + f(x_1)) = \frac{h}{2} [f(x_0) + f(x_1)]$$

$$I_{1,1} = \frac{x_2 - x_1}{2} [f(x_1) + f(x_2)] = \frac{h}{2} [f(x_1) + f(x_2)]$$

$$I_{1,2} = \frac{x_3 - x_2}{2} [f(x_2) + f(x_3)] = \frac{h}{2} [f(x_2) + f(x_3)]$$

$$I_{1,3} = \frac{x_4 - x_3}{2} [f(x_3) + f(x_4)] = \frac{h}{2} [f(x_3) + f(x_4)]$$

$$\vdots$$

$$I_{1,m-1} = \frac{x_m - x_{m-1}}{2} [f(x_{m-1}) + f(x_m)] = \frac{h}{2} [f(x_{m-1}) + f(x_m)]$$

Summing these,

$$C_{1,m} = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{m-1}) + f(x_m)]$$

Example: Find  $\int_0^2 e^x dx$  using composite Newton Cotes formula where no. of subintervals,  $m=2$  and  $m=4$

for,  $m=2$ 

$$h = \frac{b-a}{m} = \frac{2-0}{2} = 1$$

$$\begin{cases} x_0 = 0 \\ x_1 = x_0 + h = 1 \\ x_2 = 2 \end{cases}$$

$$C_{1,2} = \frac{h}{2} [f(x_0) + 2f(x_1) + f(x_2)]$$

$$= \frac{1}{2} [e^0 + 2e^1 + e^2]$$

$$= \boxed{6.9128}$$

for  $m=4$ 

$$h = \frac{2-0}{4} = \frac{1}{2}$$

$$\begin{cases} x_0 = a = 0 \\ x_1 = x_0 + h = \frac{1}{2} \\ x_2 = x_1 + h = 1 \\ x_3 = x_2 + h = \frac{3}{2} \\ x_4 = x_3 + h = 2 \end{cases}$$

$$C_{1,4} = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)]$$

$$= \frac{1/2}{2} [e^0 + 2e^{1/2} + 2e^1 + 2e^{3/2} + e^2]$$

$$= \boxed{6.5216}$$

\* The more subinterval we take, the smaller the error converges.

$$m \uparrow \quad \text{error} \downarrow$$

$$h \downarrow$$



The error decreases by a factor of 4 indicating that it is quadratic converges.

$$\int_0^2 e^x dx = [e^x]_0^2 = e^2 - e^0 = e^2 - 1 \approx 6.389$$

| m  | $G_m(f)$ | Error = $ I(f) - G_m(f) $ |
|----|----------|---------------------------|
| 1  | 8.389    | 2.000                     |
| 2  | 6.912    | 0.524                     |
| 4  | 6.522    | 0.133                     |
| 8  | 6.422    | 0.033                     |
| 16 | 6.397    | 0.008                     |

$$[(m-1)x]^2 + (m-1)x = [(m-1)x]^2 + (m-1)x = \frac{(m-1)^2 x^2}{2} = \frac{(m-1)^2}{2} x^2$$

Closed Newton Cotes formula for  $n=2$  [Simpson Rule]

$$I_2(f) = \int_a^b P_2(x) dx$$

$$P_2(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2)$$

As  $n=2$ , we need 3 data points  $x_0, x_1, x_2$

$$x_0 = a, \quad x_1 = m = \frac{a+b}{2}, \quad x_2 = b$$

$$l_0(x) = \frac{x-x_1}{x_0-x_1} \frac{x-x_2}{x_0-x_2} = \frac{x-m}{a-m} \frac{x-b}{a-b}$$

$$l_1(x) = \frac{x-x_0}{x_1-x_0} \frac{x-x_2}{x_1-x_2} = \frac{x-a}{m-a} \frac{x-b}{m-b}$$

$$l_2(x) = \frac{x-x_0}{x_2-x_0} \frac{x-x_1}{x_2-x_1} = \frac{x-a}{b-a} \frac{x-m}{b-m}$$

$$\alpha_k = \int_a^b l_k(x) dx$$

The more refined we take the smaller the error converges.

↓ error ↑ m



$$\sigma_0 = \int_a^b l_0(x) dx$$

$$= \int_a^b \frac{x-m}{a-m} \cdot \frac{x-b}{a-b} dx$$

$$= \frac{1}{(a-m)(a-b)} \int_a^b \frac{(x-m)(x-b)}{x^2 - (m+b)x + mb} dx$$

$$= \frac{1}{(a-m)(a-b)} \left[ \frac{x^3}{3} - (m+b)\frac{x^2}{2} + mbx \right]_a^b$$

$$= \frac{1}{(a-m)(a-b)} \left[ \frac{b^3}{3} - (m+b)\frac{b^2}{2} + mb^2 - \frac{a^3}{3} + (m+b)\frac{a^2}{2} - mab \right]$$

$$= \dots$$

$$\sigma_0 = \frac{b-a}{6}$$

$$\sigma_1 = \int_a^b l_1(x) dx$$

$$= \int_a^b \frac{x-a}{m-a} \frac{x-b}{m-b} dx$$

$$\vdots$$

$$\sigma_1 = \frac{2}{3}(b-a)$$

$$\sigma_2 = \int_a^b l_2(x) dx$$

$$= \int_a^b \frac{(x-a)(x-m)}{b-a} \frac{(x-m)}{b-m} dx$$

$$\sigma_2 = \frac{b-a}{6}$$

$$I_2(f) = \sum_{k=0}^2 \sigma_k f(x_k)$$

$$= \sigma_0 f(x_0) + \sigma_1 f(x_1) + \sigma_2 f(x_2)$$

$$= \frac{b-a}{6} f(a) + \frac{2}{3}(b-a) f\left(\frac{a+b}{2}\right)$$

$$+ \frac{b-a}{6} f(b)$$

$$I_2(f) = \frac{b-a}{6} \left[ f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Simpson Rule



Example: Find  $\int_0^2 (e^{0.5x} + \sin x) dx$  using closed Newton Cotes formula for  $n=2$ .

$$a=0, b=2 \quad m = \frac{a+b}{2} = \frac{0+2}{2} = 1$$

$$I_2(f) = \frac{b-a}{6} [f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)]$$

$$I_2(f) = \frac{2-0}{6} [f(0) + 4f(1) + f(2)]$$

$$= \frac{2-0}{6} (1 + 4 \times 2.4902 + 3.6276)$$

$$= 4.8628$$

The detailed steps of integration:

\* Show that  $\omega_0(x) = \frac{b-a}{6}$  for closed Newton Cotes formula for  $n=2$ .

$$\omega_0(x) = \int_a^b \omega(x) dx$$

$$= \int_a^b \frac{x-m}{a-m} \cdot \frac{x-b}{a-b} dx$$

$$= \frac{1}{(a-m)(a-b)} \int_a^b (x-m)(x-b) dx$$

$$= \frac{1}{\left(a - \frac{a+b}{2}\right)(a-b)} \int_a^b (x^2 - mx - bx + mb) dx$$

$$= \frac{1}{\left(\frac{2a-a-b}{2}\right)(a-b)} \left[ \frac{x^3}{3} - \frac{mx^2}{2} - \frac{bx^2}{2} + mbx \right]_a^b$$

$$= \frac{1}{\left(\frac{a-b}{2}\right)(a-b)} \left[ \frac{b^3}{3} - \frac{mb^2}{2} - \frac{b^3}{2} + mb^2 - \frac{a^3}{3} + \frac{ma^2}{2} - \frac{ba^2}{2} + mba \right]$$



$$= \frac{2}{(a-b)^2} \left[ \frac{2b^3 - 3mb^2 - 3b^3 + 6mb^2 - 2a^3 + 3ma^2 + 3ba^2 - 6mba}{6} \right]$$

$$= \frac{2}{(a-b)^2} \cdot \frac{1}{6} (-b^3 + 3mb^2 - 2a^3 + 3ma^2 + 3ba^2 - 6mba)$$

$$= \frac{2}{(a-b)^2} \cdot \frac{1}{6} \left[ -b^3 + 3b^2 \left( \frac{a+b}{2} \right) - 2a^3 + 3a^2 \left( \frac{a+b}{2} \right) + 3a^2b - 6ab \left( \frac{a+b}{2} \right) \right]$$

$$= \frac{1}{(a-b)^2 \times 3} \left[ -b^3 + \frac{3ab^2}{2} + \frac{3b^3}{2} - 2a^3 + \frac{3a^3}{2} + \frac{3a^2b}{2} + 3a^2b - \frac{6a^2b}{2} - \frac{6ab^2}{2} \right]$$

$$= \frac{1}{(a-b)^2 \times 3} \left[ \frac{-2b^3 + 3ab^2 + 3b^3 - 4a^3 + 3a^3 + 3a^2b + 6a^2b - 6a^2b - 6ab^2}{2} \right]$$

$$= \frac{1}{(a-b)^2 \times 3} \times \frac{b^3 - 3ab^2 - a^3 + 3a^2b}{2}$$

$$= \frac{(b-a)^3}{(b-a)^2 \times 6}$$

$$[(a-b)^2 = (b-a)^2]$$

$$\sigma_0 = \frac{b-a}{6}$$

(Showed)